

Problem 6.7

Show that the ratio of the accumulated value of 1 invested at rate i for n periods, to the accumulated value of 1 invested at rate j for n periods, where $i > j$, is equal to the accumulated value of 1 invested for n periods at rate r . Find an expression for r as a function of i and j .

The accumulated value of 1 invested at rate i for n periods is

$$(1 + i)^n.$$

The accumulated value of 1 invested at rate j for n periods is

$$(1 + j)^n.$$

Hence their ratio is

$$\frac{(1 + i)^n}{(1 + j)^n} = \left(\frac{1 + i}{1 + j} \right)^n.$$

We are told that this ratio is equal to the accumulated value of 1 invested for n periods at rate r , namely

$$(1 + r)^n.$$

So we must have

$$(1 + r)^n = \left(\frac{1 + i}{1 + j} \right)^n.$$

Taking the n th root of both sides,

$$1 + r = \frac{1 + i}{1 + j}.$$

Therefore,

$$r = \frac{1 + i}{1 + j} - 1 = \frac{i - j}{1 + j}.$$

Thus,

$$\boxed{r = \frac{i - j}{1 + j}}.$$